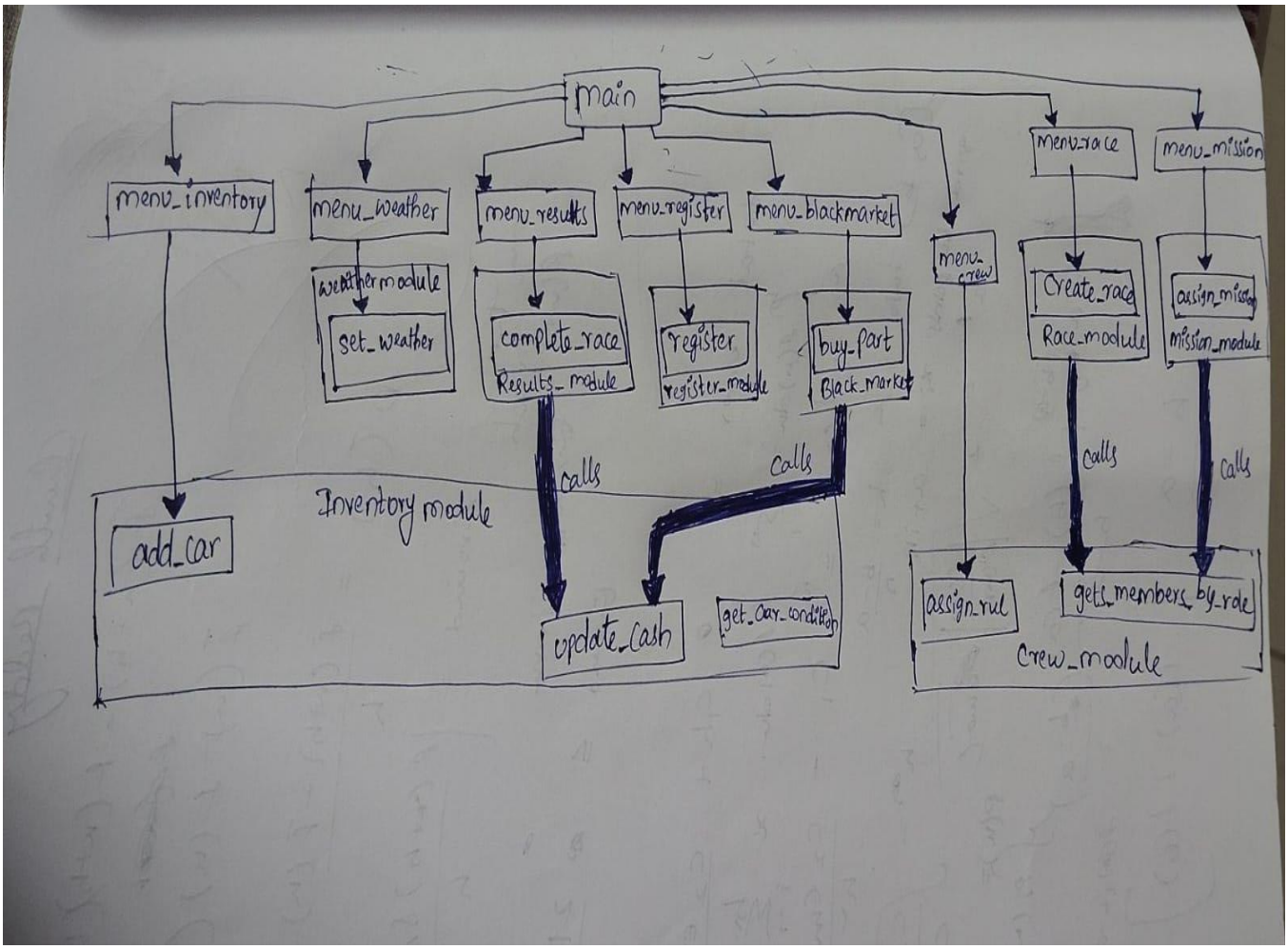
Integration Testing Report

2.1 Call Graph



Here is the step-by-step breakdown of how the integration tests flow across modules:

- `test_register_driver_and_enter_race`
  - Calls `RegistrationModule.register()`
  - Calls `CrewManagementModule.assign_role()`
  - Calls `InventoryModule.add_car()`
  - Calls `RaceManagementModule.create_race()`
    - Which interacts with `CrewManagementModule.get_members_by_role()`
    - Which interacts with `InventoryModule` to check car availability and condition.
- `test_complete_race_updates_inventory`
  - Calls `ResultsModule.complete_race()`
    - Which interacts with
    - Which modifies

deposit prize money.

to `InventoryModule.update_cash()` ndition.

- `test_assign_mission_validates_crew_roles`
  - Calls `MissionPlanningModule.assign_mission()`
    - Which interacts with `InventoryModule` to check car condition.
    - Which calls `CrewManagementModule.get_members_by_role()` to verify the needed crew member role is available and assigned.

(Please draw this manually for submission as required).

2.2 Integration Test Design

Scenario 1: Registering a driver and entering a race

- **Modules involved**: Registration, Crew Management, Inventory, Race Management.
- **Expected result**: Race is created successfully with correct driver and car.
- **Actual result**: Passed successfully. Race is in "Ready" status.
- **Logical issues found**: None.

Scenario 2: Attempting to enter race without a registered driver

- **Modules involved**: Inventory, Race Management, Crew Management.
- **Expected result**: A ValueError should be raised confirming the unregistered driver cannot be used.
- **Actual result**: Passed. ValueError: "Brian is not a valid driver for a race."

- **Logical issues found**: Handled gracefully. CrewManagement limits assignments to registered drivers, blocking invalid assignments.

Scenario 3: Completing race and verifying updates to inventory

- **Modules involved**: Race Management, Results, Inventory.
- **Expected result**: Cash balance strictly increases by the prize amount. Associated car's condition deteriorates correctly based on race completion rank.
- **Actual result**: Passed. Cash updated and condition reduced by 10 points. No errors found.

Scenario 4: Assigning a mission to damaged car without a Mechanic

- **Modules involved**: Inventory, Mission Planning, Crew Management.
- **Expected result**: Mission cannot start unless role requested is "mechanic" when the car constraint dictates it.
- **Actual result**: Passed successfully. System blocked a "driver" from attempting a mission on a severely broken car.

Scenario 5: Additional modules integration

- **Modules involved**: Weather, BlackMarket, Inventory.
- **Expected result**: BlackMarket interacts directly with Inventory deducting cash and adding parts. Weather modifies global runtime state accurately.
- **Actual result**: Passed. Validated independent interaction.

We successfully built and integrated the system ensuring accurate interactions avoiding logical dead-ends.